Business Context Structure

Current Structure

There're 20 sections and section-to-ownership is unknown (Who owns What).

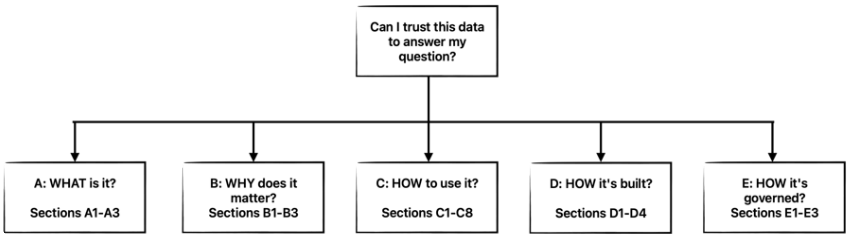
1. Table Overview
2. What This Table Is About
3. Complete Column Reference with Data Insights
4. Data Pipeline & Infrastructure
5. Primary Key & Performance
6. SLA & Refresh Schedule
7. Organizational Context & Ownership
8. Key Business Value & Use Cases
9. Key Features, Capabilities & Limitations
10. Important Notes & Pitfalls
11. Data Source Reference
12. Always-On Column Filters
13. Table Creation & Population Logic
14. Primary Use Cases
15. Best Practices & Tips
16. Common Business Metrics
17. Data Quality Checks
18. Advanced Analytics Use Cases
19. Code Implementation & ETL
20. Related Articles & Documentation

Note: For an example, please refer to Business Context - payment_cogs_audit

Future Structure

The Governing Thought

A business context document must answer one question: **"Can I trust this data asset to answer my business question correctly?"**



To answer "yes," a reader needs five things — in this order:

1. **What is it?** — Identity, purpose, and who owns it
2. **What questions does it answer?** — Value, use cases, and relevance to my problem
3. **How do I use it correctly?** — Schema, rules, metrics, glossary, queries, and pitfalls
4. **How is it built?** — Sources, pipeline, schedule, and transformation logic
5. **How is it governed?** — Quality checks, standards, and related documentation

The Recommended Structure: 5 Pillars, 20 Sections

- For an example, please refer to Business Context v2 - payment_cogs_audit

Pillar A: WHAT Is It? — Identity & Purpose

> **Governing question:** "In 30 seconds, do I know what this table is, what one row represents, and who is responsible for it?"

#	Section	Description	Content Guidance

A1	**Table Overview**	The identity card — names, schemas, URLs, IDs, and grain	Compact key-value table. Must include **grain** (what one row represents), database, schema, catalog links, approximate row count, and date range.
A2	**What This Table Is About**	Plain-language narrative of the business process captured	2-3 paragraphs. What business process? What core analysis does it enable? What time period and scope? No ETL details — those belong in Pillar D.
A3	**Organizational Context & Ownership**	Who owns, maintains, and is accountable for this data asset	Owner team, steward, escalation path, domain. Critical for knowing who to ask when something looks wrong.

Pillar B: WHY Does It Matter? — Value & Use Cases

> **Governing question:** "Does this table answer my specific question, and what has it been used for?"

#	Section	Description	Content Guidance
B1	**Key Business Value**	The business problems this table was built to solve	State the value proposition: what decisions does this data inform? What would be impossible without it? Pure strategic value — no specific questions here.
B2	**Primary Use Cases**	Specific, recurring analytical questions this table answers	Bullet list of questions in natural language (e.g., "What is the monthly COGS variance by merchant?"). This is the bridge that lets AI agents match a user's question to the right table.
B3	**Advanced Analytics Use Cases**	Complex or cross-table analytical scenarios	Multi-table joins, cohort analyses, trend detection, anomaly identification. May be empty for simple tables.

Pillar C: HOW Do I Use It Correctly? — Schema, Rules & Guidance

> **Governing question:** "What do I need to know to write correct SQL and interpret results without making mistakes?"

This is the largest pillar because it directly determines whether the answer to a business question is **right or wrong**. It has 8 sections — 40% of the document — reflecting that SQL correctness and data interpretation are the highest-risk steps in the analytical workflow.

#	Section	Description	Content Guidance
C1	**Complete Column Reference with Data Insights**	Full schema with types, descriptions, sample values, and statistics	One table for readability. Include lineage (where each column comes from), category, and key statistics. This is the most-referenced section for SQL development.
C2	**Primary Key & Performance**	Key constraints, partitioning, and query optimization	Primary key (or explicit statement that none exists and how to deduplicate). Partition columns. Always-filter-on guidance. Index information if relevant.
C3	**Key Features, Capabilities & Limitations**	What the data can and cannot do	Capabilities (dimensions available, time range, granularity). Limitations (missing processors, no customer-level detail, known gaps).
C4	**Important Notes & Pitfalls**	Gotchas that lead to wrong answers if missed	Sign conventions, null handling, edge cases, outlier warnings, implicit filters. The "read this or get burned" section.
C5	**Always-On Column Filters**	Filters permanently applied during ETL that restrict the data scope	e.g., processor_name = 'L' only. Critical because queries cannot override these — the data simply isn't there.
C6	**Common Business Metrics**	Standardized metric definitions and SQL formulas	Table of metric name, formula, and reporting grain. Ensures consistent reporting across teams. Include sign conventions and NULLIF guards.

C7	**Glossary & Term Definitions**	Domain-specific terms, abbreviations, and conventions used in this table	Table of term, definition, and context. Covers business terms (COGS, interchange), code values (processor 'L' = Elavon), and data conventions (actual_cogs_amt is negative). The #1 preventer of silent misinterpretation by both humans and AI agents.
C8	**Example Queries & Patterns**	Concrete, working SQL examples demonstrating correct usage	2-5 annotated queries covering common use cases. Each query implicitly encodes grain, sign conventions, partition filters, join keys, and metric formulas. AI agents learn correct patterns from examples more reliably than from rules.

Pillar D: HOW Is It Built? — Pipeline & Provenance

> **Governing question:** "Where does the data come from, how is it transformed, and when is it available?"

#	Section	Description	Content Guidance
D1	**Data Source Reference**	Upstream source systems and tables feeding this asset	Source system names, source tables, extraction method. The authoritative answer to "where does this data originate?"
D2	**Data Pipeline & Infrastructure**	Repository, code paths, and infrastructure	Repo name, file paths, orchestration tool (Airflow DAG, etc.), compute platform.
D3	**SLA & Refresh Schedule**	When data is available and what it depends on	Delivery cadence, dependency tables, timing (e.g., "billing data arrives on the 1st of the following month"). Latency expectations.
D4	**Table Creation & ETL Implementation**	The query/process that builds the table and detailed ETL logic	Reference to creation query, key transformations, join logic, process flow, error handling, schema migration. Use subsections to organize: Creation Logic, Process Flow, Data Processing Steps, Error Handling. For complex tables, summarize rather than embedding hundreds of lines of SQL.

Pillar E: HOW Is It Governed? — Quality, Standards & Ecosystem

> **Governing question:** "Can I trust the data quality, and where do I find more context?"

#	Section	Description	Content Guidance
E1	**Data Quality Checks**	Automated and manual validation rules	List each check with: what it validates, when it runs, what happens on failure. Include suggested ad-hoc quality queries.
E2	**Best Practices & Tips**	Practical guidance for querying and interpreting the data	Query performance tips, common mistakes to avoid, interpretation guidance. Complements "Important Notes & Pitfalls" with positive guidance rather than warnings.
E3	**Related Articles & Documentation**	Links to external resources, related tables, and cross-references	Catalog links, wiki pages, related data assets, domain documentation. The "see also" section.

Authoring / Ownership Priority

#	Section	Owner	Reviewer(s)	Metadata Source(s)		Notes
				Primary	**Secondary**	
	Pillar A: WHAT Is It?	**DP**				@Shobhit Nagpal could you please fill out the columns of Metadata Source(s) so that DE & BA can update as needed.
A1	Table Overview	**DP**	DE	Alation	Lineage, GitHub	
A2	What This Table Is About	**BA**	DP, DG	Alation	Lineage, GitHub,	

					Table Data Sampler	
A3	Organizational Context & Ownership	**DG**	DP, BA	Alation	Lineage, GitHub	
	Pillar B: WHY Does It Matter?	**BA**				
B1	Key Business Value	**BA**	DP, DG	Alation	—	
B2	Primary Use Cases	**BA**	DG	Alation	—	
B3	Advanced Analytics Use Cases	**BA**	DE, DG	Alation	—	
	Pillar C: HOW Do I Use It Correctly?					
C1	Complete Column Reference with Data Insights	**DE + BA**	DP, DG	Alation, GitHub	Table Data Sampler	
C2	Primary Key & Performance	**DE**	DG	Alation	GitHub	
C3	Key Features, Capabilities & Limitations	**BA + DE**	DP, DG	Alation	GitHub	
C4	Important Notes & Pitfalls	**DE + BA**	DP, DG	Alation	—	
C5	Always-On Column Filters	**DE**	BA, DG	Alation	—	

C 6	Common Business Metrics	**BA**	DE, DP, DG	Alation	—	
C 7	Glossary & Term Definitions	**BA + DE**	DP, DG			
C 8	Example Queries & Patterns	**BA + DE**	DG			
	Pillar D: HOW Is It Built?					
D1	Data Source Reference	**DP**	DE, DG	Lineage	—	
D 2	Data Pipeline & Infrastructure	**DE**	DP	Lineage	GitHub, Alation	
D 3	SLA & Refresh Schedule	**DE**	DP, BA	GitHub	Alation	
D 4	Table Creation & ETL Implementation	**DE**	DP, DG			
	Pillar E: HOW Is It Governed?					
E1	Data Quality Checks	**DG + DE**	DP, BA	GitHub	Alation, Table Data Sampler	
E 2	Best Practices & Tips	**BA + DE**	DG	Alation	GitHub	
E 3	Related Articles & Documentation	**DG**	BA	Alation	—	